

A Negative Answer to the Universal-Function Version of Erdős's Third Question in Problem 514

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Abstract

Let f be a transcendental entire function and let

$$M_f(r) = \max_{|z|=r} |f(z)|.$$

Erdős asked, among other things, whether there exists a path to infinity along which $|f(z)|$ tends to infinity faster than a fixed function of $M_f(|z|)$, for instance faster than a positive power of $M_f(|z|)$. This note addresses only that third question, in the natural universal sense that the comparison function is fixed in advance and is independent of f . We prove a negative answer in a strong form. For every nondecreasing function $\Phi(T) \rightarrow \infty$ there is a transcendental entire function f such that every path to infinity γ satisfies

$$\liminf_{t \rightarrow \infty} \frac{|f(\gamma(t))|}{\Phi(M_f(|\gamma(t)|))} = 0.$$

Consequently no universal comparison function, even one growing much more slowly than every positive power, can be forced in the universal-comparison interpretation of Erdős's third question. The result follows from a theorem of Hayman on the growth of entire functions on asymptotic paths, after a short selection argument for Hayman's auxiliary growth function. We also include a direct construction, which gives a compact and formalization-friendly alternative proof of the same theorem.

1 The formulation addressed here

Let f be a transcendental entire function, that is, an entire function which is not a polynomial. For $r > 0$, put

$$M_f(r) := \max_{|z|=r} |f(z)|.$$

A path to infinity is a continuous map

$$\gamma : [0, \infty) \rightarrow \mathbb{C}$$

such that

$$|\gamma(t)| \rightarrow \infty \quad (t \rightarrow \infty).$$

Erdős's third question in Problem 514 asks whether one can force $|f(z)|$ to tend to infinity along a path faster than a fixed function of $M_f(r)$; see [1, p. 249] and the later problem list [2]. We formalize the relevant positive assertion as follows.

Problem 1 (Universal comparison-function formulation). *Does there exist a function $\Psi : [T_0, \infty) \rightarrow (0, \infty)$, with $\Psi(T) \rightarrow \infty$ as $T \rightarrow \infty$, independent of f , such that every transcendental entire function f has a path to infinity γ for which*

$$\frac{|f(\gamma(t))|}{\Psi(M_f(|\gamma(t)|))} \rightarrow \infty \quad (t \rightarrow \infty)?$$

All statements involving $\Psi(M_f(r))$ or $\Phi(M_f(r))$ are understood for sufficiently large r , so that $M_f(r) \geq T_0$. This is harmless, since $M_f(r) \rightarrow \infty$ for every nonconstant entire function.

It is no loss to restrict attention to nondecreasing comparison functions. Indeed, suppose that a not-necessarily-monotone function $\Psi : [T_0, \infty) \rightarrow (0, \infty)$, with $\Psi(T) \rightarrow \infty$, were to work. Choose $T_1 \geq T_0$ so large that $\Psi(S) \geq 1$ for all $S \geq T_1$, and on $[T_1, \infty)$ define the lower envelope

$$\Phi(T) := \inf_{S \geq T} \Psi(S).$$

Then $1 \leq \Phi \leq \Psi$, the function Φ is nondecreasing, and $\Phi(T) \rightarrow \infty$. The last assertion follows because, for each $C > 0$, we have $\Psi(S) \geq C$ for all sufficiently large S . A path with $|f(\gamma(t))|/\Psi(M_f(|\gamma(t)|)) \rightarrow \infty$ would then also have $|f(\gamma(t))|/\Phi(M_f(|\gamma(t)|)) \rightarrow \infty$. Thus it suffices to rule out nondecreasing Φ 's; discarding the initial interval $[T_0, T_1)$ is irrelevant for all asymptotic statements.

Related work. The first two questions in Problem 514 are closely related to the subharmonic-function path results of Huber, Lewis–Rossi–Weitsman, and Wu. Chojecki's note [4] applies this line to obtain the existence and length assertions. For the parenthetical power-type part of the third question, Chojecki used Langley's formulation [7, Theorem 1.4] of a consequence of Barth–Brannan–Hayman [3]. There is also a complementary finite-lower-order/multiple-tract line: Rossi and Weitsman [9] proved a power lower bound in $|z|$ along an asymptotic path when a high-level set has at least two components, and Toda [10] sharpened this type of result, including a subharmonic analogue. These results are positive path-growth theorems in terms of $|z|$ and tract geometry, rather than universal lower bounds in terms of $M_f(r)$. The present note is concerned only with the universal-comparison formulation; for that formulation Hayman's theorem gives the result after Lemma 1, and the direct construction below is a self-contained and formalization-friendly proof of the same corollary.

We prove the following stronger negative statement.

Theorem 1. *Let*

$$\Phi : [T_0, \infty) \rightarrow (0, \infty)$$

be nondecreasing and satisfy $\Phi(T) \rightarrow \infty$ as $T \rightarrow \infty$. Then there exists a transcendental entire function f such that for every path to infinity γ ,

$$\liminf_{t \rightarrow \infty} \frac{|f(\gamma(t))|}{\Phi(M_f(|\gamma(t)|))} = 0.$$

Since the quotient is nonnegative whenever it is defined, the conclusion is equivalently the following epsilon form: for every path γ , every $\eta > 0$, and every $T_ \geq 0$, there exists $t \geq T_*$, with $M_f(|\gamma(t)|) \geq T_0$, such that*

$$\frac{|f(\gamma(t))|}{\Phi(M_f(|\gamma(t)|))} \leq \eta.$$

Corollary 1. *There is no universal function $\Psi(T) \rightarrow \infty$, independent of f , for which every transcendental entire function has a path to infinity along which*

$$|f(z)|/\Psi(M_f(|z|)) \rightarrow \infty.$$

In particular, the parenthetical power-type case $\Psi(T) = T^\varepsilon$, $\varepsilon > 0$, also fails.

2 Hayman's theorem and the selection step

The main input is Hayman's theorem on the growth of entire functions on asymptotic paths. We use "integral function" in Hayman's classical sense, namely entire function. A path Γ is an asymptotic path of f if $z \rightarrow \infty$ along Γ and $f(z) \rightarrow \infty$ along Γ .

Theorem 2 (Hayman [5, Theorem 2, p. 253]). *Let $\lambda(r)$ be a positive increasing function of $r > 0$ such that*

$$\frac{\log \lambda(r)}{\log r} \rightarrow \infty \quad (r \rightarrow \infty).$$

Then there exist two entire functions f_1 and f_2 , each of infinite lower order, such that, for all sufficiently large r ,

$$\log M(r, f_1) < \exp(\lambda(r)) \quad \text{and} \quad \log M(r, f_2) > \exp(\lambda(r)).$$

Moreover, for $j = 1, 2$, if Γ_j is an asymptotic path of f_j , then

$$\limsup_{\substack{z \rightarrow \infty \\ z \in \Gamma_j}} \frac{\log \log |f_j(z)|}{\log |z|} < \infty.$$

Only the fast-growth function f_2 is needed below.

Lemma 1 (Selection of Hayman's auxiliary growth). *Let $\Phi : [T_0, \infty) \rightarrow (0, \infty)$ be nondecreasing and satisfy $\Phi(T) \rightarrow \infty$ as $T \rightarrow \infty$. There exists a positive increasing function $\lambda(r)$, $r > 0$, such that*

$$\frac{\log \lambda(r)}{\log r} \rightarrow \infty$$

and, for every positive integer k ,

$$\Phi(\exp(\exp(\lambda(r)))) \geq \exp(r^k)$$

for all sufficiently large r .

Proof. For each integer $n \geq 2$, choose $S_n \geq \max\{T_0, e\}$ such that

$$\Phi(S_n) \geq \exp(n^n),$$

which is possible because $\Phi(T) \rightarrow \infty$. Choose a strictly increasing sequence a_n , $n \geq 2$, such that

$$a_n \geq n^n, \quad a_n \geq \log \log S_n, \quad \exp(\exp(a_n)) \geq S_n.$$

Define $\lambda(n) = a_n$ for integers $n \geq 2$, interpolate linearly on each interval $[n, n+1]$, and extend λ on $(0, 2]$ so that it is positive and increasing on $(0, \infty)$.

If $n \leq r \leq n+1$, then $\lambda(r) \geq a_n$, and hence

$$\frac{\log \lambda(r)}{\log r} \geq \frac{\log a_n}{\log(n+1)} \geq \frac{\log(n^n)}{\log(n+1)} \rightarrow \infty.$$

Thus λ has Hayman's required growth.

Fix a positive integer k . For $n \leq r \leq n+1$, monotonicity of Φ and the choices above give

$$\Phi(\exp(\exp(\lambda(r)))) \geq \Phi(\exp(\exp(a_n))) \geq \Phi(S_n) \geq \exp(n^n).$$

For all sufficiently large n , $n^n \geq r^k$ throughout $[n, n+1]$. Therefore

$$\Phi(\exp(\exp(\lambda(r)))) \geq \exp(r^k)$$

for all sufficiently large r , as required. $\square$

Proof of Theorem 1 from Hayman's theorem. Choose λ as in Lemma 1, and let $f = f_2$ be the fast-growth entire function supplied by Theorem 2. Since f has infinite lower order, it is transcendental entire. Hayman's maximum modulus conclusion gives, for all sufficiently large r ,

$$M_f(r) > \exp(\exp(\lambda(r))).$$

Let $\gamma : [0, \infty) \rightarrow \mathbb{C}$ be any path to infinity. Recall that, by our definition, this means that γ is continuous and $|\gamma(t)| \rightarrow \infty$. If γ is not an asymptotic path of f , then $|f(\gamma(t))|$ does not tend to infinity. Hence there are a constant $B > 0$ and a sequence $t_m \rightarrow \infty$ such that

$$|f(\gamma(t_m))| \leq B$$

for all m . Since f is nonconstant entire, $M_f(r) \rightarrow \infty$, and hence

$$\Phi(M_f(|\gamma(t_m)|)) \rightarrow \infty.$$

Along this sequence the quotient in Theorem 1 tends to zero.

It remains to consider the case where γ is an asymptotic path of f . Let $\Gamma = \gamma([0, \infty))$. The compactness of $\gamma([0, T])$ for each fixed T implies that estimates holding as $z \rightarrow \infty$ along Γ hold eventually as $t \rightarrow \infty$ along γ . Hayman's path estimate therefore gives a finite constant C such that, eventually along γ , with $r = |\gamma(t)|$,

$$\log \log |f(\gamma(t))| \leq C \log r.$$

Equivalently, after increasing C harmlessly and discarding an initial segment of the path,

$$|f(\gamma(t))| \leq \exp(r^C).$$

Choose an integer $k > C$. By the fast-growth bound for M_f , monotonicity of Φ , and Lemma 1,

$$\Phi(M_f(r)) \geq \Phi(\exp(\exp(\lambda(r)))) \geq \exp(r^k)$$

eventually along the path. Therefore

$$0 \leq \frac{|f(\gamma(t))|}{\Phi(M_f(|\gamma(t)|))} \leq \exp(r^C - r^k) \rightarrow 0.$$

In particular the required liminf is zero. Since γ was arbitrary, Theorem 1 follows. $\square$

3 A direct alternative construction

The proof above shows that Theorem 1 is a corollary of Hayman's Theorem 2 plus the elementary selection lemma. We now give a self-contained construction proving the same theorem directly. The point of this second proof is not priority for the theorem statement, but rather a short construction whose estimates are explicit and well suited to formal checking.

Fix Φ as in Theorem 1. Put

$$\varepsilon_n := (-1)^{n+1}, \quad n = 1, 2, 3, \dots$$

Thus $\varepsilon_n = 1$ for odd n and $\varepsilon_n = -1$ for even n .

Choose

$$x_0 > 0$$

so large that $e^x/2 \geq T_0$ whenever $x \geq x_0$. For $x \geq x_0$, define

$$A(x) := \min \left\{ \Phi(e^x/2), \frac{e^x}{4} \right\}.$$

Since both $\Phi(e^x/2)$ and $e^x/4$ tend to infinity, we have

$$A(x) \rightarrow \infty \quad (x \rightarrow \infty).$$

We shall construct positive real numbers

$$r_n, \quad X_n, \quad \lambda_n, \quad a_n$$

with $r_n \rightarrow \infty$, $X_n \rightarrow \infty$, and

$$a_n = \lambda_n r_n - X_n.$$

The entire function will be

$$f(z) := \sum_{n=1}^{\infty} \exp(-a_n + \varepsilon_n \lambda_n z).$$

The parameters are chosen so that the following off-diagonal estimates hold for all $m \neq n$:

$$\exp(-a_m + \lambda_m r_n) \leq 2^{-m-n-4} A(X_n). \quad (1)$$

We also require

$$A(X_n) \geq 2^{n+4}, \quad X_n \geq n \log r_n + n, \quad X_n \geq x_0, \quad (2)$$

and

$$a_n \geq n^2 + \lambda_n n. \quad (3)$$

Lemma 2. *The sequences r_n, X_n, λ_n, a_n can be chosen so that (1), (2), and (3) hold.*

Proof. We argue by induction. Suppose that the parameters have already been chosen for indices $< n$. Choose

$$r_n > \max\{r_{n-1}, n\},$$

with the convention that there is no restriction involving r_0 when $n = 1$.

The quantities

$$\exp(-a_m + \lambda_m r_n), \quad m < n,$$

are now fixed and finite. Since $A(x) \rightarrow \infty$, we may choose X_n so large that (2) holds and, for every $m < n$,

$$\exp(-a_m + \lambda_m r_n) \leq 2^{-m-n-4} A(X_n).$$

This gives the off-diagonal estimates in which already constructed terms are evaluated at the new radius r_n .

It remains to make the new term small at all previous radii. For each $k < n$, the inequality

$$\exp(X_n - \lambda_n(r_n - r_k)) \leq 2^{-n-k-4} A(X_k)$$

will hold once λ_n is sufficiently large, because $r_n - r_k > 0$. Since there are only finitely many such k , one value of λ_n can make all these inequalities true. Increasing λ_n further if necessary, we also ensure

$$\lambda_n(r_n - n) \geq X_n + n^2,$$

which is possible because $r_n > n$. Now set

$$a_n := \lambda_n r_n - X_n.$$

Then

$$a_n = \lambda_n r_n - X_n \geq n^2 + \lambda_n n,$$

so (3) holds, and for $k < n$,

$$\exp(-a_n + \lambda_n r_k) = \exp(X_n - \lambda_n(r_n - r_k)) \leq 2^{-n-k-4} A(X_k).$$

Thus all required estimates involving index n and previous indices hold. The induction is complete. $\square$

4 The function is transcendental entire

Let

$$f(z) = \sum_{n=1}^{\infty} \exp(-a_n + \varepsilon_n \lambda_n z)$$

be the function constructed above.

First we show that the series converges locally uniformly. Fix $R > 0$ and suppose $|z| \leq R$. If $n > R$, then by (3),

$$-a_n + \lambda_n |z| \leq -a_n + \lambda_n R \leq -a_n + \lambda_n n \leq -n^2.$$

Hence

$$|\exp(-a_n + \varepsilon_n \lambda_n z)| \leq e^{-n^2}$$

for all sufficiently large n , uniformly on $|z| \leq R$. Therefore the series converges uniformly on compact subsets of $\mathbb{C}$, and f is entire.

Next we show that f is not a polynomial. At the point

$$z_n := \varepsilon_n r_n, \quad |z_n| = r_n,$$

the n -th term equals

$$\exp(-a_n + \varepsilon_n \lambda_n z_n) = \exp(-a_n + \lambda_n r_n) = e^{X_n}.$$

For the other terms, (1) gives

$$\sum_{m \neq n} |\exp(-a_m + \varepsilon_m \lambda_m z_n)| \leq \sum_{m \neq n} \exp(-a_m + \lambda_m r_n) \leq 2^{-n-4} A(X_n).$$

Since $A(X_n) \leq e^{X_n}/4$, this is at most $e^{X_n}/16$. Consequently,

$$M_f(r_n) \geq |f(z_n)| \geq \frac{1}{2} e^{X_n}. \quad (4)$$

Together with (2), this gives

$$M_f(r_n) \geq \frac{1}{2} e^{X_n} \geq \frac{1}{2} e^n r_n^n.$$

If f were a polynomial of degree d , then $M_f(r) = O(r^d)$, which is incompatible with the last estimate along the sequence r_n as $n \rightarrow \infty$. Thus f is transcendental entire.

5 Smallness on the wrong half-plane

Fix n , and let $|z| = r_n$. We call the half-plane

$$\varepsilon_n \operatorname{Re} z > 0$$

the correct half-plane for the n -th circle. Thus the correct half-plane is the right half-plane for odd n and the left half-plane for even n .

Suppose $|z| = r_n$ and $\varepsilon_n \operatorname{Re} z \leq 0$. Then the main term satisfies

$$|\exp(-a_n + \varepsilon_n \lambda_n z)| = \exp(-a_n + \varepsilon_n \lambda_n \operatorname{Re} z) \leq e^{-a_n} \leq 1.$$

The other terms are bounded by

$$\sum_{m \neq n} |\exp(-a_m + \varepsilon_m \lambda_m z)| \leq \sum_{m \neq n} \exp(-a_m + \lambda_m r_n) \leq 2^{-n-4} A(X_n).$$

By (2), $A(X_n) \geq 2^{n+4}$, and hence

$$1 \leq 2^{-n-4} A(X_n).$$

Therefore

$$|f(z)| \leq 2^{-n-3} A(X_n).$$

On the other hand, from (4) and the monotonicity of Φ ,

$$\Phi(M_f(r_n)) \geq \Phi(e^{X_n}/2) \geq A(X_n),$$

where $e^{X_n}/2 \geq T_0$ follows from $X_n \geq x_0$. We have proved the key estimate

$$|z| = r_n, \quad \varepsilon_n \operatorname{Re} z \leq 0 \quad \implies \quad |f(z)| \leq 2^{-n-3} \Phi(M_f(r_n)). \quad (5)$$

6 Smallness on the imaginary axis

For any real y ,

$$|\exp(-a_n + \varepsilon_n \lambda_n i y)| = e^{-a_n}.$$

Thus, by (3),

$$|f(iy)| \leq \sum_{n=1}^{\infty} e^{-a_n} \leq \sum_{n=1}^{\infty} e^{-n^2} =: B < \infty.$$

Since f is transcendental entire, in particular nonconstant, $M_f(r) \rightarrow \infty$. Hence $\Phi(M_f(r)) \rightarrow \infty$. It follows that

$$\frac{|f(iy)|}{\Phi(M_f(|y|))} \rightarrow 0 \quad (|y| \rightarrow \infty), \quad (6)$$

where the quotient is understood for sufficiently large $|y|$, so that $M_f(|y|) \geq T_0$.

7 Every path has small values infinitely often

Let $\gamma : [0, \infty) \rightarrow \mathbb{C}$ be any path to infinity. Since $r_n \rightarrow \infty$, for every sufficiently large n the path meets the circle $|z| = r_n$. Let

$$\tau_n := \inf\{t \geq 0 : |\gamma(t)| = r_n\}$$

be the first hitting time. For all large n , τ_n is finite, $\tau_n < \tau_{n+1}$, and $\tau_n \rightarrow \infty$. These facts follow from the continuity of $|\gamma(t)|$ and the condition $|\gamma(t)| \rightarrow \infty$.

There are two cases.

First suppose that for infinitely many n ,

$$\varepsilon_n \operatorname{Re} \gamma(\tau_n) \leq 0.$$

For those n , estimate (5) gives

$$\frac{|f(\gamma(\tau_n))|}{\Phi(M_f(|\gamma(\tau_n)|))} \leq 2^{-n-3} \rightarrow 0.$$

Thus the desired liminf is zero.

It remains to consider the opposite case. Then, for all sufficiently large n ,

$$\varepsilon_n \operatorname{Re} \gamma(\tau_n) > 0.$$

Since ε_n changes sign with n , the real parts $\operatorname{Re} \gamma(\tau_n)$ and $\operatorname{Re} \gamma(\tau_{n+1})$ have opposite signs for all large n . By continuity of γ , there exists

$$s_n \in [\tau_n, \tau_{n+1}]$$

such that

$$\operatorname{Re} \gamma(s_n) = 0.$$

Write $\gamma(s_n) = iy_n$, where $y_n \in \mathbb{R}$. Since $s_n \geq \tau_n \rightarrow \infty$ and $|\gamma(t)| \rightarrow \infty$, we have $|y_n| = |\gamma(s_n)| \rightarrow \infty$. Hence (6) gives

$$\frac{|f(\gamma(s_n))|}{\Phi(M_f(|\gamma(s_n)|))} = \frac{|f(iy_n)|}{\Phi(M_f(|y_n|))} \rightarrow 0.$$

Again the desired liminf is zero.

Since γ was arbitrary, Theorem 1 follows.

8 Consequences and formalization note

Theorem 1 proves Corollary 1 as follows. Let $\Psi(T) \rightarrow \infty$ be any proposed fixed comparison function. If Ψ is nondecreasing, applying Theorem 1 directly with $\Phi = \Psi$ produces a transcendental entire function f for which every path to infinity has

$$\liminf_{t \rightarrow \infty} \frac{|f(\gamma(t))|}{\Psi(M_f(|\gamma(t)|))} = 0.$$

For a nonmonotone Ψ , apply Theorem 1 instead to the nondecreasing lower envelope Φ constructed in the first section. Since $\Phi \leq \Psi$ on the relevant tail, any sequence of points for which the quotient with denominator $\Phi(M_f(|z|))$ tends to zero also has the quotient with denominator $\Psi(M_f(|z|))$ tending to zero. Hence no path can make the Ψ -quotient tend to infinity. This rules out the universal comparison-function version of the third question in Erdős Problem 514.

The note does not reprove the first two questions in Problem 514, nor does it claim priority for the parenthetical power-type case discussed in the related work above. The direct construction above is self-contained and proves the same negative statement, while the earlier argument records the shorter derivation from Hayman’s theorem. The accompanying Lean 4 file [Erdos514.lean](#), using `leanprover/lean4:v4.28.0` and the corresponding `mathlib` version, formalizes the theorem in the equivalent epsilon form. In that file the main circle-maximum theorem is `comparison_negative_sphere`; the predicate `FrequentlySmall` is the epsilon formulation above; and the circle maximum `Ms` matches $M_f(r) = \max_{|z|=r} |f(z)|$. The Lean proof first establishes a closed-disk version, `comparison_negative_closedDisk`, and then identifies it with the circle version by the maximum-modulus principle. The Lean file treats Φ as a total function $\mathbb{R} \rightarrow \mathbb{R}$, with monotonicity and positivity assumed only on $[T_0, \infty)$; this is equivalent to the partial-domain formulation used here after extending Φ arbitrarily outside the relevant tail.

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